

HDOC DAY-29

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Batch-15

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QUESTION:

Write a C program to perform menu-driven operations on arrays such as counting even, odd, positive, negative and zero elements, searching an element using linear search, merging two arrays, and exiting the program.

Aim

To write a C program to perform various array operations using a menu-driven approach.

Algorithm

1. Start the program.
2. Display the menu of array operations.
3. Ask the user to enter a choice.
4. Perform the selected operation.
5. Display the result.
6. Repeat the menu until the user selects **Exit**.
7. Stop.

Input

- Size and elements of the array.
- User's menu choice.
- Element to be searched.
- Elements of the second array for merging.

Output

- Count of even, odd, positive, negative, and zero elements.
- Position of the searched element, if found.
- Merged array after combining two arrays.

Variables Used

- `a[]` – stores elements of first array
- `b[]` – stores elements of second array
- `n` – size of first array
- `m` – size of second array
- `choice` – stores menu choice
- `key` – element to be searched
- `i` – loop variable

TEST CASE:

Test Case	Operation	Input	Expected Output
1	Count elements	2 -3 0 4 -5	Even = 3, Odd = 2, Positive = 2, Negative = 2, Zero = 1
2	Linear Search	10 20 30 40, Search = 30	Element found at position 3
3	Merge Arrays	1 2 3 and 4 5	1 2 3 4 5

>C program:

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int a[10], b[10];
6     int n, m, i, ch, x;
7     int even, odd, pos, neg, zero;
8
9     do
10    {
11        printf("\n1. Count");
12        printf("\n2. Search");
13        printf("\n3. Merge");
14        printf("\n4. Exit");
15        printf("\nEnter choice: ");
16        scanf("%d", &ch);
17
18        if(ch == 1)
19        {
20            printf("Enter size: ");
21            scanf("%d", &n);
22
23            for(i=0; i<n; i++)
24                scanf("%d", &a[i]);
25
26            even=odd=pos=neg=zero=0;
27
28            for(i=0; i<n; i++)
29            {
30                if(a[i]<=0)
31                    even++;
32                else
33                    odd++;
34
35                if(a[i]>0)
36                    pos++;
37                else if(a[i]<0)
38                    neg++;
39                else
40                    zero++;
41            }
42
43            printf("Even = %d\n", even);
44            printf("Odd = %d\n", odd);
45            printf("Positive = %d\n", pos);
46            printf("Negative = %d\n", neg);
47            printf("Zero = %d\n", zero);
48        }
49        else if(ch == 2)
50        {
51            printf("Enter size: ");
52

```

```
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- day29.c - Mousepad

50 else if(ch == 2)
51 {
52     printf("Enter size: ");
53     scanf("%d", &n);
54     for(i=0; i<n; i++)
55     {
56         scanf("%d", &a[i]);
57     }
58     printf("Enter number to search: ");
59     scanf("%d", &x);
60     for(i=0; i<n; i++)
61     {
62         if(a[i] == x)
63         {
64             printf("Found at position %d\n", i+1);
65             break;
66         }
67     }
68     if(i == n)
69     {
70         printf("Not found\n");
71     }
72 }
73
74 else if(ch == 3)
75 {
76     printf("Enter first array size: ");
77     scanf("%d", &n);
78     for(i=0; i<n; i++)
79     {
80         scanf("%d", &a[i]);
81     }
82     printf("Enter second array size: ");
83     scanf("%d", &m);
84     for(i=0; i<n; i++)
85     {
86         scanf("%d", &b[i]);
87     }
88     printf("Merged array: ");
89     for(i=0; i<n; i++)
90     {
91         printf("%d ", a[i]);
92     }
93     for(i=0; i<m; i++)
94     {
95         printf("%d ", b[i]);
96     }
97 } while(ch != 4);
98
99 return 0;
100 }
101
```

>Output:

```
Session Actions Edit View Help
priyadarshna@kali: ~
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali]~$ gcc day29.c -o day29
[priyadarshna@kali]~$ ./day29
1. Count
2. Search
3. Merge
4. Exit
Enter choice: 1
Enter size: 5
2 3 4 6 7
Even = 3
Odd = 2
Positive = 3
Negative = 1
Zero = 1
1. Count
2. Search
3. Merge
4. Exit
Enter choice: 2
Enter size: 5
10 20 30 40 50
Enter number to search: 30
Found at position 3
1. Count
2. Search
3. Merge
4. Exit
Enter choice: 3
Enter first array size: 3
1 2 3
Enter second array size: 3
4 5 6
Merged array: 1 2 3 4 5 6
1. Count
2. Search
3. Merge
4. Exit
Enter choice: 4
[priyadarshna@kali]~$
```

